Chapter: Basic Concepts

Kinematics of Fluid in Motion

1.1 Introduction

Fluids include gases and liquids, with air and water being the most common examples. Fluid mechanics is involved in many aspects of life, such as flow in pipelines and channels, movement of air and blood, drag forces, wind loading on buildings, motion of projectiles, lubrication, irrigation, sedimentation, meteorology, and oceanography. It is also essential in designing water supply systems, wastewater treatment facilities, aircraft, ships, submarines, rockets, turbines, pumps, and heating and air-conditioning systems.

Fluid mechanics is more complex than solid mechanics because fluids have no distinct elements. It is generally divided into three branches: (i) Fluid Statics, which studies fluids at rest; (ii) Kinematics, which deals with fluid motion without considering forces or energy; and (iii) Fluid Dynamics, which studies the relationship between motion and the forces acting on fluids. Hydrodynamics refers specifically to the study of moving incompressible fluids.

Fluids are classified into two types: (i) Liquids, which are nearly incompressible and experience little volume change with pressure, and (ii) Gases, which are compressible and change volume when pressure varies. Although hydrodynamics originally focused on ideal frictionless fluids, developments in aeronautics and chemical engineering led to the broader field of fluid mechanics, which studies the behavior of real liquids and gases.

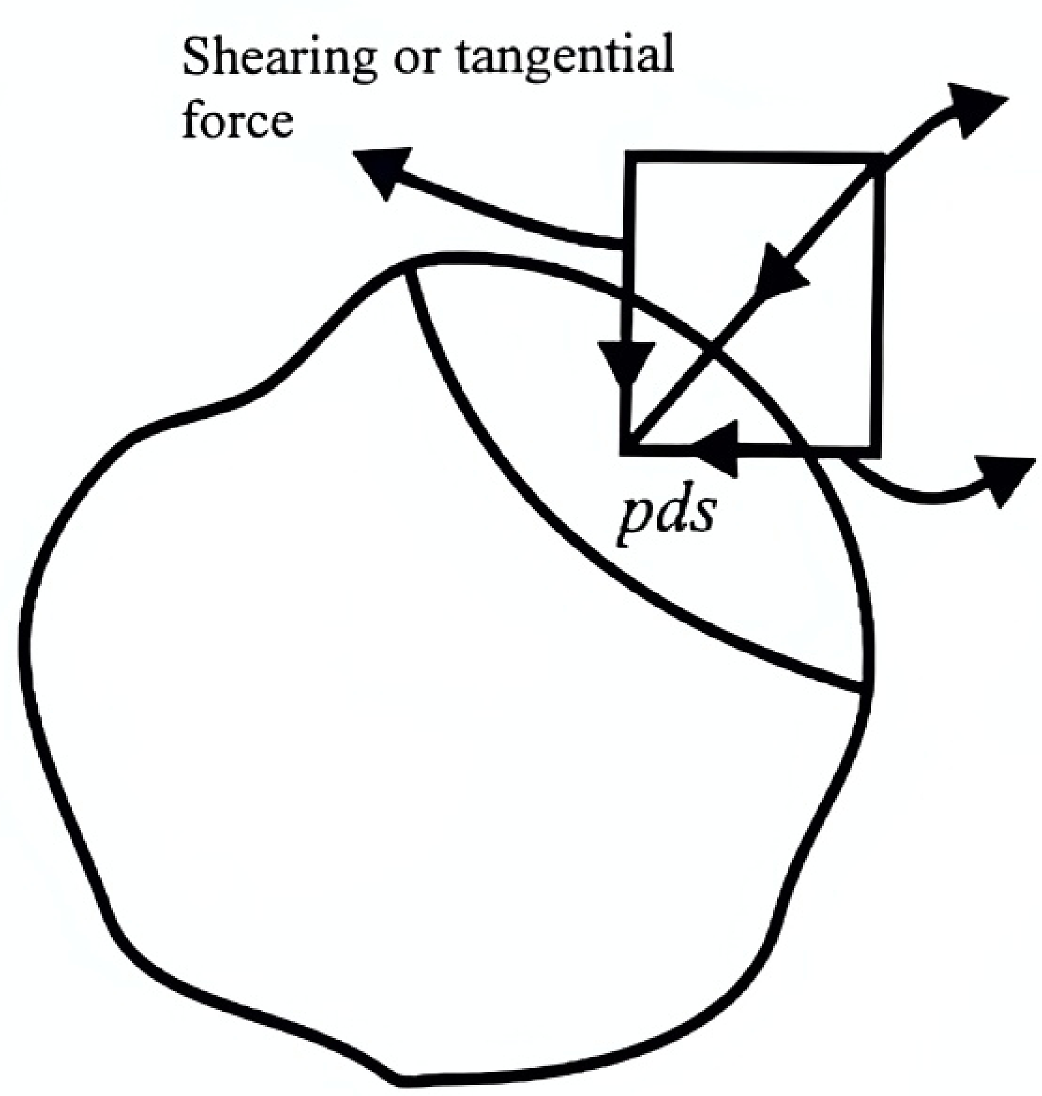
On the microscopic scale, matter consists of molecules in random motion separated by varying distances. However, in fluid mechanics, fluids are treated as continuous media with unique properties such as velocity, pressure, and density at every point. An infinitesimal volume of fluid is called a fluid particle.

Important contributions to fluid mechanics were made by Leonardo da Vinci (1452-1519) through experimental studies of fluid flow and by Isaac Newton (1642-1727) through his laws of motion and viscosity. During the 17th and 18th centuries, many frictionless flow problems were solved, though inviscid-flow solutions were often found inadequate for describing real fluids.

1.2 Real and Ideal Fluid

An infinitesimal fluid element is acted upon by two types of forces, namely body forces and surface forces. The former is a type of force, which is proportional to the mass (or possible the volume) of the body on which it acts while the later is one, which acts on a surface element, and it is proportional to the surface area.

Suppose that the fluid element is enclosed by the surface S. Let P be an arbitrary point on S and let ds be the surface element around P. Then surface force on ds is in general not in the direction of normal at P to ds.



Hence the force may result into components, one normal and the other tangential with the area ds. The normal force per unit area is said to be normal stress or pressure while the tangential force per unit area is said to be shearing or tangential stress.

A fluid is said to be real (or viscous) when normal as well as shearing stress exists. On the other hand, a fluid is said to be ideal (or perfect or non-viscous or frictionless or inviscid) when it does not exert any shearing stress, whether at rest or in motion. Water and air are treated as ideal fluid whereas syrup and heavy oil are treated as viscous fluid.

1.3 Some Important Types of Flow

i) Laminar (Stream line) and Turbulent Flow

A flow in which each particle traces out a definite curve and the curves traced out by any two different fluid particles do not intersect is said to be Laminar.

On the other hand, a flow in which each fluid particle does not trace out a definite curve and the curves traced out by fluid particles intersect is said to be turbulent.

ii) Steady and Unsteady Flow

A flow in which properties and conditions associated with the motion of the fluid are independent of time so that the flow pattern remains unchanged with time is said to be steady.

On the other hand, a flow in which properties and conditions associated with the motion of the fluid depends on time so that the flow pattern varies with time is said to be unsteady.

iii) Uniform and Non-uniform Flows

A flow in which the fluid particles possess equal velocities at each section of the channel or pipe is called uniform, while if the fluid particles possess different velocities at each section, the flow is called non-uniform.

iv) Rotational and Irrotational Flows

A flow in which the particles of the fluid go on rotating about their own axis while flowing are said to be rotational.

On the other hand, if the fluid particle S does not rotate about their own axis, the flow is said to be irrotational.

v) Barotropic flow

A flow is aid to be barotropic when the pressure is a function of density.

1.4 Method of Describing Fluid Motion

There are two methods for studying fluid motion mathematically viz. Lagrangian and Eulerian method and refer to "individual time-rate of change" and "local time-rate change" respectively.

Eulerian Method

In this method, we select any point fixed in space occupied by the fluid and observe the changes which take place in velocity, density, and pressure as the fluid passes through this point. Obviously, the point being fixed x, y, z and t are independent variables and so $\dot{x}, \ddot{x}$ etc are meaning less.

We consider any scalar point function $\varphi(x,y,z,t)$ or $\varphi(\vec{r},t)$ associated with a fluid in motion, and then keeping the point (x,y,z) as fixed, the change is $\varphi(\vec{r},t+\delta t)-\varphi(\vec{r},t)$.

Therefore $\lim\limits_{\delta t\to 0}\dfrac{\varphi(\vec{r},t+\delta t)-\varphi(\vec{r},t)}{\delta t}=\dfrac{\partial\varphi}{\partial t}$ is the local time rate of change.

A similar expression can be established for a vector point function $\vec{f}(\vec{r},t)$ and

$$\lim_{\delta t\to 0}\frac{\vec{f}(\vec{r},t+\delta t)-\vec{f}(\vec{r},t)}{\delta t}=\frac{\partial\vec{f}}{\partial t}$$

1.5 Orthogonal Curvilinear Coordinates

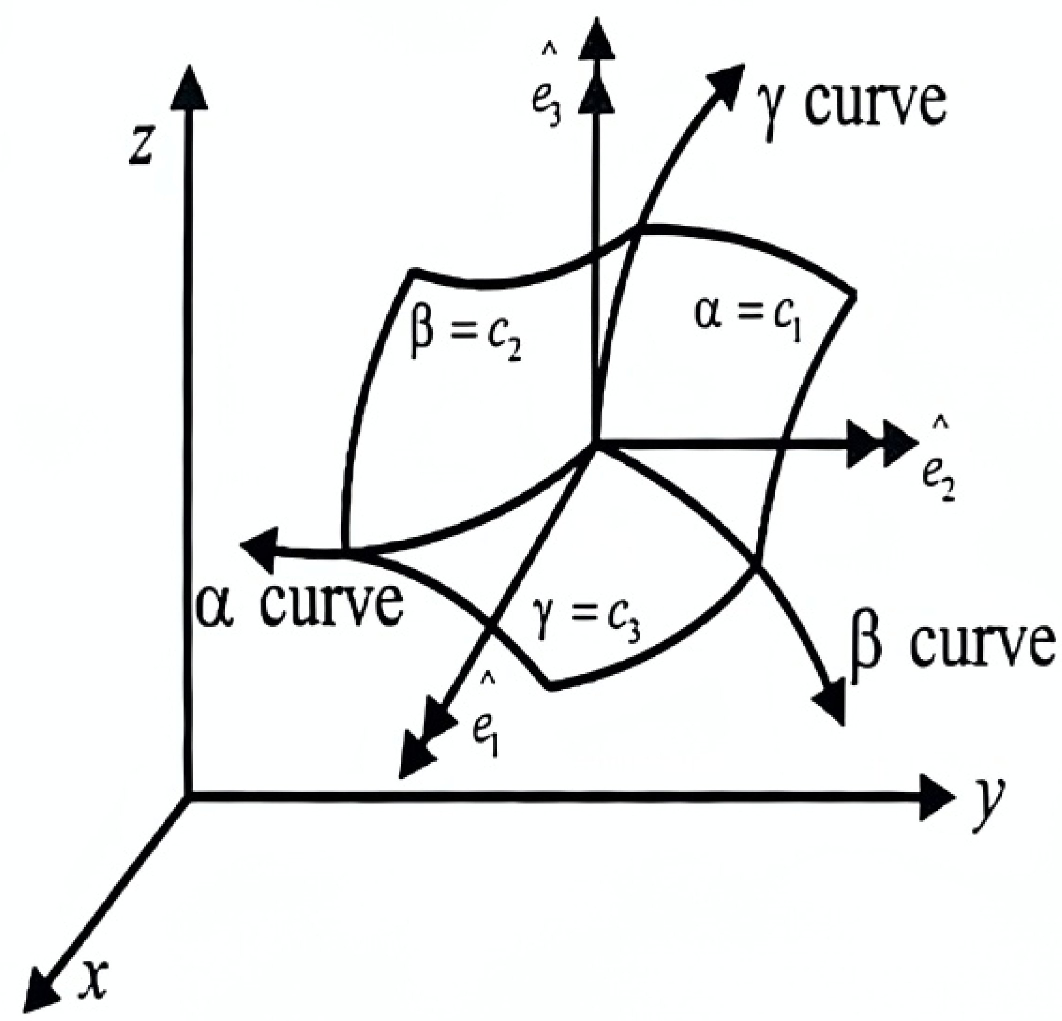
Consider three orthogonal families of surfaces $f_1(x,y,z)=\alpha$, $f_2(x,y,z)=\beta$, $f_3(x,y,z)=\gamma$ (1)

where (x,y,z) are the Cartesian coordinates. The surfaces $\alpha=\text{constant}$, $\beta=\text{constant}$ and $\gamma=\text{constant}$ for an orthogonal system and the values of α,β,γ may be used as the coordinates (orthogonal curvilinear coordinates of a point in surface).

The relation between the two systems of coordinates (x,y,z) and (α,β,γ) may be expressed by equation (1).

or, $x=x(\alpha,\beta,\gamma)$, $y=y(\alpha,\beta,\gamma)$ and $z=z(\alpha,\beta,\gamma)$ (2)

The surfaces $\alpha=c_1$, $\beta=c_2$, $\gamma=c_3$ where c_1,c_2,c_3 are constants are called coordinate surfaces.



Let, $\vec{r}=\hat{i}x+\hat{j}y+\hat{k}z$ be the position vector of a point P. Then equation (2) can be written as $\vec{r}=\vec{r}(\alpha,\beta,\gamma)$.

A tangent vector to the α-curve at P (for which β=constant and γ=constant) is $\dfrac{\partial\vec{r}}{\partial\alpha}$.

Hence a unit tangent vector in this direction is

$$\hat{e}_1=\frac{\dfrac{\partial\vec{r}}{\partial\alpha}}{\left|\dfrac{\partial\vec{r}}{\partial\alpha}\right|}\Rightarrow\frac{\partial\vec{r}}{\partial\alpha}=h_1\hat{e}_1$$

where $h_1=\left|\dfrac{\partial\vec{r}}{\partial\alpha}\right|=\sqrt{\left(\dfrac{\partial x}{\partial\alpha}\right)^2+\left(\dfrac{\partial y}{\partial\alpha}\right)^2+\left(\dfrac{\partial z}{\partial\alpha}\right)^2}$ (3)

Similarly, the unit vectors $\hat{e}_2$ and $\hat{e}_3$ to the β-curve and γ-curve are given by $\dfrac{\partial\vec{r}}{\partial\beta}=h_2\hat{e}_2$ and $\dfrac{\partial\vec{r}}{\partial\gamma}=h_3\hat{e}_3$.

The quantities h_1,h_2,h_3 are called scale factors.

Now, $\vec{r}=\vec{r}(\alpha,\beta,\gamma)$, we get $d\vec{r} = \frac{\partial \vec{r}}{\partial \alpha} d\alpha + \frac{\partial \vec{r}}{\partial \beta} d\beta + \frac{\partial \vec{r}}{\partial \gamma} d\gamma = h_1 d\alpha \hat{e}_1 + h_2 d\beta \hat{e}_2 + h_3 d\gamma \hat{e}_3$ (4)

Thus, $ds^2 = d\vec{r} \cdot d\vec{r} = h_1^2 d\alpha^2 + h_2^2 d\beta^2 + h_3^2 d\gamma^2$ (5)

Also, along α-curve, β-curve and γ-curves are constants. So that $dr_1 = h_1 d\alpha \hat{e}_1$ where $ds_1 = h_1 d\alpha$. Similarly, the differential arc length along β-curve and γ-curve at P are $ds_2 = h_2 d\alpha$, $ds_3 = h_3 d\alpha$

If then we draw the surfaces corresponding $(\alpha, \alpha + d\alpha), (\beta, \beta + d\beta), (\gamma, \gamma + d\gamma)$, we obtain an elementary equilibrium parallelepiped whose edges are $h_1 d\alpha, h_2 d\beta, h_3 d\gamma$. We shall require the following vector definitions: for any scalar φ and for any vector $\vec{q} = (q_1, q_2, q_3)$ we have,

$$grad\varphi = \nabla\varphi = \left(\frac{1}{h_1}\frac{\partial \varphi}{\partial \alpha} + \frac{1}{h_2}\frac{\partial \varphi}{\partial \beta} + \frac{1}{h_3}\frac{\partial \varphi}{\partial \gamma} \right)$$

$$div\vec{q} = \vec{\nabla}.\vec{q} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial \alpha}(h_1 h_3 q_1) + \frac{\partial}{\partial \beta}(h_3 h_1 q_2) + \frac{\partial}{\partial \gamma}(h_1 h_2 q_3) \right]$$

$$\vec{w} = curl\vec{q} = \vec{\nabla} \times \vec{q} = (w_1, w_2, w_3)$$

where $w_1 = \frac{1}{h_2 h_3}\left[\frac{\partial}{\partial \beta}(h_3 q_3) - \frac{\partial}{\partial \gamma}(h_2 q_2) \right]$

$$w_2 = \frac{1}{h_3 h_1}\left[\frac{\partial}{\partial \gamma}(h_2 q_2) - \frac{\partial}{\partial \alpha}(h_1 q_1) \right]$$

$$w_3 = \frac{1}{h_1 h_2}\left[\frac{\partial}{\partial \alpha}(h_1 q_1) - \frac{\partial}{\partial \beta}(h_3 q_3) \right]$$

$$\nabla^2\varphi = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial \alpha}\left(\frac{h_2 h_3}{h_1}\frac{\partial \varphi}{\partial \alpha} \right) + \frac{\partial}{\partial \beta}\left(\frac{h_3 h_1}{h_2}\frac{\partial \varphi}{\partial \beta} \right) + \frac{\partial}{\partial \gamma}\left(\frac{h_1 h_2}{h_3}\frac{\partial \varphi}{\partial \gamma} \right) \right] \qquad (6)$$

Particular Case

A. Cartesian Coordinates (x, y, z)

Here $ds^2 = dx^2 + dy^2 + dz^2$ $\therefore$ Surfaces factors $h_1 = h_2 = h_3 = 1$

Volume element $= dxdydz$

$$grad\varphi = \nabla\varphi = \left(\frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} + \frac{\partial \varphi}{\partial z} \right)$$

$$div\vec{q} = \vec{\nabla}.\vec{q} = \left(\frac{\partial q_1}{\partial x} + \frac{\partial q_2}{\partial y} + \frac{\partial q_3}{\partial z}\right)$$

$$\vec{w} = curl\vec{q} = \vec{\nabla}\times\vec{q} = (w_1, w_2, w_3)$$

where $w_1 = \frac{\partial q_3}{\partial y} - \frac{\partial q_2}{\partial z}$, $w_2 = \frac{\partial q_1}{\partial z} - \frac{\partial q_3}{\partial x}$, $w_3 = \frac{\partial q_2}{\partial x} - \frac{\partial q_1}{\partial y}$

and $\nabla^2\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2}$ (7)

Let $\vec{r}$ be the position of a fluid particle at a time t, so that $\vec{q}$ is the velocity of the particle at that point then, $\vec{q} = \vec{q}(\vec{r}, t)$.

Suppose that the velocity at time $t + \delta t$ when the particle has moved to a neighboring position $\vec{r} + \delta\vec{r}$ is $\vec{q} + \delta\vec{q}$, then $\delta\vec{q} = \vec{q}(\vec{r} + \delta\vec{r}, t + \delta t) - \vec{q}(\vec{r}, t)$

$$= \left\{\vec{q}(\vec{r} + \delta\vec{r}, t + \delta t) - \vec{q}(\vec{r}, t + \delta t)\right\} + \left\{\vec{q}(\vec{r}, t + \delta t) - \vec{q}(\vec{r}, t)\right\}$$

$$= (\delta\vec{r}\cdot\vec{\nabla})\vec{q}(\vec{r}, t + \delta t) + \delta t\frac{\partial\vec{q}(\vec{r}, t)}{\partial t} \tag{8}$$

Dividing both side by δt and making we get, $\frac{D\vec{q}}{Dt} = \left(\frac{D\vec{r}}{Dt}\cdot\vec{\nabla}\right)\vec{q}(\vec{r}, t) + \frac{\partial\vec{q}(\vec{r}, t)}{\partial t}$ (9)

i.e., $\frac{D\vec{q}}{Dt} = (\vec{q}\cdot\vec{\nabla})\vec{q} + \frac{\partial\vec{q}}{\partial t}$ (10)

$$= \vec{\nabla}\left(\frac{1}{2}q^2\right) + \vec{w}\wedge\vec{q} + \frac{\partial\vec{q}}{\partial t} \tag{11}$$

where, $\vec{w} = curl\vec{q} = \vec{\nabla}\wedge\vec{q}$

The expressions (10) or (11) is known as Lagrangian acceleration and the vector $\vec{w}\wedge\vec{q}$ is called Lamb vector.

Particular Cases

Cartesian Coordinates (x, y, z)

Let u, v, w be the velocity components in the x, y, z directions. Then noting that $\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$, $q^2 = u^2 + v^2 + w^2$

$$\vec{w}=\left(\frac{\partial w}{\partial y}-\frac{\partial v}{\partial z},\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x},\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right) \tag{12}$$

The acceleration components f_x, f_y, f_z are given from (11) as,

$$f_x=\frac{Dx}{Dt}=\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z} \tag{13}$$

and two similar relations for f_y and f_z.

1.6 Equation of Continuity or (Conservation of mass)

The law of conservation of mass states that fluid mass can neither be created nor can be destroyed. The equation of continuity expresses the law of conservation of mass in mathematical form. Thus, in continuous motion, the equation of continuity expresses the fact that the increase in the mass of the fluid within any closed surface drawn in the fluid must be equal to the excess of the mass that flows in over the mass that flows out.

Equation of Continuity (Vector Form) by Euler's Method

Let, S be an arbitrary small closed surface drawn in the fluid enclosing a volume V and let S be taken fixed in space. Let P be any point of S and P be the fluid density at P at any time t.

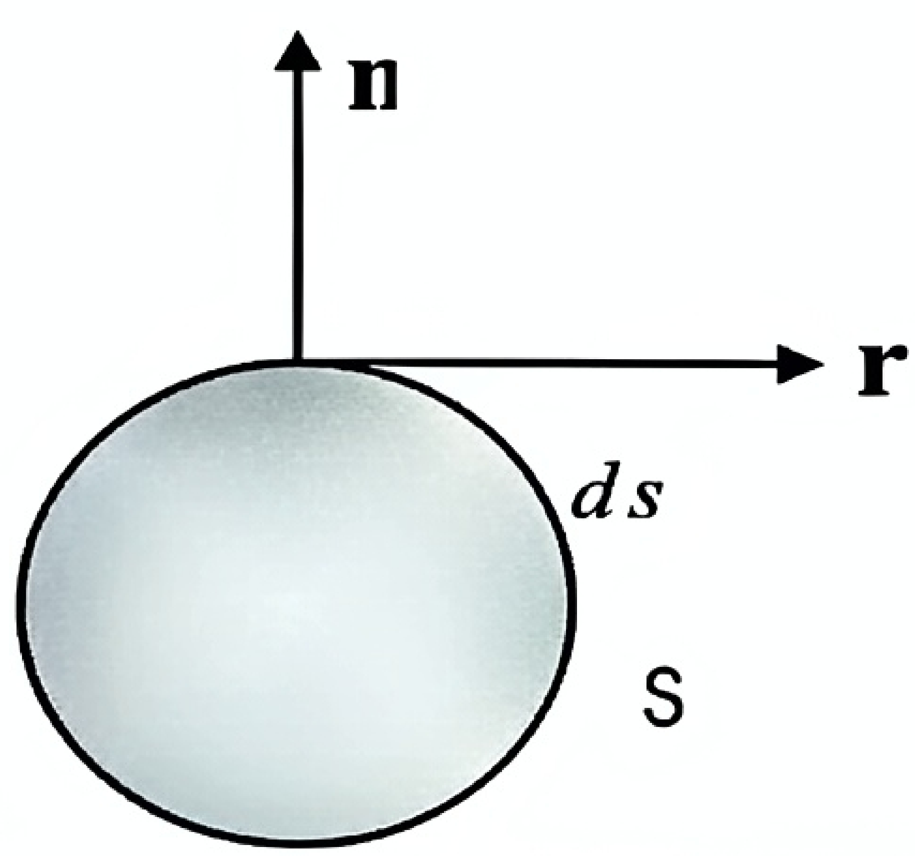


Figure 1. Geometrical configuration of equation of continuity

Let δs denote element of the surface S enclosed in P, $\hat{n}$ being the outward drawn unit normal at δs and $\vec{q}$ be the fluid velocity at P. Then the normal component of $\vec{q}$ measured outward from V is $\vec{q}\cdot\hat{n}$.

Rate of mass flow across $\delta s=\rho\vec{q}\cdot\hat{n}\delta s$

$\therefore$ The total rate of mass flow across S$=\int_s \rho(\vec{q}\cdot\hat{n})ds=\int_V \vec{\nabla}\cdot(\rho\vec{q})dv$

$\therefore$ The total rate of mass flow into V $= -\int_V \vec{\nabla}\cdot\left(\rho\vec{q}\right)dv$ (14)

Again, the mass of the fluid within S at time t $= \int_v \rho dv$

$\therefore$ The total rate of mass increase within S is $\frac{\partial}{\partial t}\left[\int_V \rho dv\right] = \int_V \frac{\partial \rho}{\partial t}dv$

Now suppose that the region V of the fluid contains neither sources nor sinks (i.e., there are no inlets or outlets through which fluid can enter or leave the region). Then by the law of conservation of fluid mass, the rate of increase of mass of the fluid within V must be equal to the total rate of mass flow into V. Hence, we have, $\int_v \frac{\partial \rho}{\partial t}dv = -\int_V \vec{\nabla}\cdot\left(\rho\vec{q}\right)dv$

$$\int_v \left\{\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\left(\rho\vec{q}\right)\right\}dv = 0 \qquad (15)$$

Since this hold for arbitrary chosen volume V we have, $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\left(\rho\vec{q}\right) = 0$ (16)

Equation (16) is called the equation of continuity or conservation of mass.

Equation (16) can be written as $\frac{\partial \rho}{\partial t} + \rho\vec{\nabla}\cdot\vec{q} + (\vec{\nabla}\rho)\cdot\vec{q} = 0$ or $\frac{D\rho}{Dt} + \rho\vec{\nabla}\cdot\vec{q} = 0$

$\frac{\partial \rho}{\partial t} + q.(\nabla\rho) + \rho(\nabla.q) = 0$

$\frac{D\rho}{Dt} + \rho\nabla.q = 0$

Corollary-1: For incompressible and heterogeneous fluid, the density is invariable with time. So that, $\frac{D\rho}{Dt} = 0$ and the equation of continuity is, $\vec{\nabla}\cdot\vec{q} = 0$

Corollary-2: For an incompressible and homogeneous fluid ρ is constant and so (16) gives $\vec{\nabla}\cdot\vec{q} = 0$.

$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + q.\nabla\rho = \left[\frac{\partial}{\partial t} + (q.\nabla)\right]\rho$

Particular Cases:

Case-1: Cartesian Coordinates (x, y, z)

$$\text{or, } \frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z} + \rho\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) = 0$$

$$\text{or, } \frac{\partial \rho}{\partial t} + \rho\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) = 0 \qquad (17)$$

If the fluid be homogeneous and incompressible, then ρ is constant and the equation of continuity is $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$

Further if the fluid is heterogeneous and incompressible then, $\frac{D\rho}{Dt}=0$

and the equation of continuity is, $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$

1.7 Stream Lines

A streamline is a curve drawn in the fluid such that at any time t, the direction of the tangent at any point of the curve coincides with the direction of the velocity of the fluid particle at that point. Thus if (u,v,w) be the velocity components of the fluid particle at $P(x,y,z)$, the direction the direction ratios of the tangent being, $d\vec{r}=(\partial x,\partial y,\partial z)$ at that point the differential equation of the stream line are $\vec{q}\wedge d\vec{r}=0$

$$\frac{\partial x}{u}=\frac{\partial y}{v}=\frac{\partial z}{w} \tag{18}$$

where, $\vec{q}=\hat{i}u+\hat{j}v+\hat{k}w$

Path Lines

A path line is a curve, which a particular fluid particle describes during its motion. The differential equation of the path lines are, $\dot{x}=u$, $\dot{y}=v$, $\dot{z}=w$ (19)

where, the $(\cdot)$ denotes the differentiation with respect to time.

Stream Tube

The stream lines drawn through each point of a closed curve enclose a tubular surface in the fluid called a stream tube or tube of flow. A tube of infinitesimal cross-section is called stream filament.

Consider a stream filament of liquid is in steady motion. Let the cross-sectional area of the filament be so small that the velocity is the same at each point of this area, which may be taken to be perpendicular to the direction of the velocity. Let v_1 and v_2 be the velocity of the fluid at cross-sections S_1 and S_2 respectively and the fluid is supposed to be incompressible. Then by the law of conservation of mass, the total quality of the liquid flowing across each section of the filament must be the same i.e., $v_1S_1=v_2S_2$. Thus, we have the result.

"The product of the velocity and the cross-sectional area is constant along a stream filament of a liquid in steady motion."

1.8 Velocity Potential

Suppose that the fluid velocity at time t be $\vec{q}=(u,v,w)$. Further suppose that at time t, there exist a scalar function $\varphi(x,y,z,t)$ uniform through out the entire fluid flow such that $udx+vdy+wdz$ is an exact differential of $-d\varphi$, so that

$$u=-\frac{\partial\varphi}{\partial x},\ v=-\frac{\partial\varphi}{\partial y},\ w=-\frac{\partial\varphi}{\partial z} \tag{20}$$

The function φ is called the velocity potential or velocity function.

The necessary and sufficient condition for (20) to hold is

$\vec{\nabla}\times\vec{q}=0$ [Since, $\vec{\nabla}\times\vec{q}=-\vec{\nabla}\times\vec{\nabla}\varphi=-curl(grad\varphi)=0$]

i.e., $\hat{i}\left(\frac{\partial w}{\partial y}-\frac{\partial v}{\partial z}\right)+\hat{j}\left(\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x}\right)+\hat{k}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=0$

So that, $\frac{\partial w}{\partial y}=\frac{\partial v}{\partial z},\ \frac{\partial u}{\partial z}=\frac{\partial w}{\partial x},\ \frac{\partial v}{\partial x}=\frac{\partial u}{\partial y}$ (21)

The surface, $\varphi(x,y,z,t)=\text{constant}$ (22)

is called the equipotential. The stream line $\frac{\partial x}{u}=\frac{\partial y}{v}=\frac{\partial z}{w}$ are cut at right angles by the surface given by differential equation, $udx+vdy+wdz=0$ (23)

And the condition for the orthogonal surface is the condition that (23) may posses a solution of the form equation (22) at the instant t, the analytic condition being given by,

$$u\left(\frac{\partial w}{\partial y}-\frac{\partial v}{\partial z}\right)+v\left(\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x}\right)+w\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=0 \tag{24}$$

For incompressible fluid φ satisfies Laplace equation, $\nabla^2\varphi=0$ (25)

1.9 Irrotational and Rotational Motion

Vorticity

If $\vec{q}$ be the velocity vector of a fluid particle, then the vector quantity

$\vec{w}=\frac{1}{2}\vec{\nabla}\times\vec{q}=\frac{1}{2}curl\vec{q}$ is called the Vorticity vector or simply Vorticity and is measured of the angular velocity of an infinitesimal element.

The component of spin ξ,η,ρ are given by

$$\xi = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right),\ \eta = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right),\ \rho = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

Vortex Line

A vortex line is a curve in the fluid such that the tangent to it at each point gives the direction of velocity $\vec{w}$ at that point. The vortex line is designated by $w-$line.

The definition of vortex line implies that, $d\vec{r} \wedge \vec{w} = 0$

$$\text{i.e.,} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ dx & dy & dz \\ \xi & \eta & \rho \end{vmatrix} = 0 \text{ i.e.,} \frac{dx}{\xi} = \frac{dy}{\eta} = \frac{dz}{\rho} \tag{26}$$

Vortex Tube

The vortex lines drawn through each point of a closed curve enclose a tubular space in the fluid called vortex tube.

A vortex tubes if infinitesimal cross-section is called a vortex filament or simply a vortex.

Irrotational and Rotational Motion

The motion of a fluid is said to be irrotational when the vorticity $\vec{w}$ of every fluid particle is zero, i.e., $\xi = \eta = \rho = 0$.

It is obvious that the velocity potential φ exists when the motion is irrotational. When the velocity $\vec{w}$ is different from zero, the motion is called rotational motion.